Algebraic Functional Equations

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1 References

There are three excellent resources on functional equation.

- Functions in Intermediate Notes
- Notes written by Evan Chen, they can be found on his personal website: web.evanchen.cc. They are Monsters and Introduction to Functional Equations. The subject also appear in Chapter 3-4 of his book, OTIS Excerpt.
- Functional Equation, a book by Pang Cheng Wu, IMO 2015 Silver Medalist. It is freely availbe on AOPS.

2 Review of Basic Knowledge

- Basic substitutions: Substituting constants, setting variables to be zero, setting variables to be equal.
- Cauchy Equations: If the domain of the function is $\mathbb{Q}$ then we instantly obtain the answer once we solved the Cauchy Equation
- Injectivity and Surjectivity

3 Four Ideas for substitutions

The main reference for this section is OTIS Excerpt Chapter 3.4

3.1 Forced Cancellation

Suppose that the equation is of the form

$$f(A) + P = f(B) + Q$$

We substitute such that $A = B$, then we get $P = Q$.

Example 1 Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + y) = f(x^{27} + 2y) + f(x^4)$$

3.2 The fff trick

If we have some expression relating $f(f(x))$ to something in terms of $f(x)$ and x, then we can replace x by $f(x)$.

Example 2 Find all strictly increasing functions $f:\mathbb{Z}\to\mathbb{Z}$ such that $f(f(x))=x+2$ for all integers x.

3.3 Symmetry

If the equation is largely symmetric w.r.t. x and y, we can swap x and y to get most of the terms cancelled.

Example 3 Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$xf(x)+y^2+f(xy)=f(x+y)^2-f(x)f(y)$$

for all real numbers x and y.

3.4 Isolated Parts

If some parts are isolated, we can use it to prove injectivity and surjectivity.

Example 4 (Kyrgyzstan MO 2012 Problem 4) Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f(f(x)^2+f(y))=xf(x)+y$$

for any $x,y\in\mathbb{R}$.

4 Transformation into Cauchy Equation

Reference: Pang Cheng Wu Chapter 2

We can make certain transformations to transform equations into Cauchy equations (See Wu P.29)

Condition	Transformation
$f(x+y)=f(x)f(y)$	$g(x)=\log f(x)$
$f(xy)=f(x)f(y)$	$g(x)=\log f(a^x)$
$f(xy)=f(x)+f(y)$	$g(x)=f(a^x)$
$f(x+y)=f(x)+f(y)+a$	$g(x)=f(x)+a$
$f(x+y+a)=f(x)+f(y)$	$g(x)=f(x+a)$
$f(x+y+a)=f(x)+f(y)+b$	$g(x)=f(x+a)+b$

5 Elementary Problems

1. (Elementary Problems 1.25) Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

   $$f(2f(f(x))+y)=2f(x+3)+f(3y-1)$$

   for any $x,y\in\mathbb{R}$.

2. (Elementary Problems 1.26) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x+y)-x)f(f(x+y)-y) = xy$$

for any $x, y \in \mathbb{R}$.

3. (Elementary Problems 1.31) Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f(x+y) + 2xy = f(x) + f(y)$.

4. (Elementary Problems 1.33) Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that

$$f(2f(x)+f(y)) = 2x+y$$

5. (Elementary Problems 1.53) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(z)+y) = zf(x)+y$$

for any $x, y, z \in \mathbb{R}$.

6. (Elementary Problems 1.54) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(y+zf(x)) = f(y)+xf(z)$$

6 More Advanced Problems

(5M) (IMO shortlist 2008 A1/ Problem 4) Find all functions $f : (0,\infty) \mapsto (0,\infty)$ (so f is a function from the positive real numbers) such that

$$\frac{(f(w))^2+(f(x))^2}{f(y^2)+f(z^2)} = \frac{w^2+x^2}{y^2+z^2}$$

for all positive real numbers w, x, y, z, satisfying $wx = yz$.

(5M) (IMO shortlist 2010 A1/ Problem 1) Find all function $f : \mathbb{R} \to \mathbb{R}$ such that for all $x, y \in \mathbb{R}$ the following equality holds

$$f(\lfloor x \rfloor y) = f(x) \lfloor f(y) \rfloor$$

where $\lfloor a \rfloor$ is greatest integer not greater than a.

(10M) (IMO shortlist 2002 A1) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)+y) = 2x + f(f(y)-x)$$

for all real x, y.

(15M) (IMO shortlist 2002 A4) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$(f(x)+f(z))(f(y)+f(t)) = f(xy-zt)+f(xt+yz)$$

for all real x, y, z, t.

(15M) (IMO shortlist 2011 A3) Determine all pairs of functions $f : \mathbb{R} \to \mathbb{R}$, $g : \mathbb{R} \to \mathbb{R}$ satisfying

$$g(f(x+y)) = f(x) + (2x+y)g(y)$$

for all real numbers x and y.

(20M) (IMO shortlist 2005 A2) Find all functions $f : \mathbb{R}^{+} \to \mathbb{R}^{+}$ such that

$$f(x)f(y) = 2f(x + yf(x))$$

for all positive real numbers x and y.

(20M) (IMo shortlist 2007 A4) Find all functions $f : \mathbb{R}^{+} \to \mathbb{R}^{+}$ such that

$$f(x + f(y)) = f(x + y) + f(y)$$

for all positive reals x and y.

(25M) (IMO shortlist 2012 A5) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y) + f(x)f(y) = f(xy) + 2xy + 1$$

for all real numbers x and y.

(30M) (IMO shortlist 2009 A7) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(x + y)) = f(yf(x)) + x^2$$

for all $x, y \in \mathbb{R}$.

(35M) (IMO shortlist 2020 A6) Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ satisfying

$$f^{a^2+b^2}(a + b) = af(a) + bf(b)$$

for all integers a and b

(40M) (IMO shortlist 2010 A6) Suppose that f and g are two functions defined on the set of positive integers and taking positive integer values. Suppose also that the equations $f(g(n)) = f(n) + 1$ and $g(f(n)) = g(n) + 1$ hold for all positive integer. Prove that $f(n) = g(n)$ for all positive integer n.

(40M) (IMO shortlist 2020 A8) Let R^{+} be the set of positive real numbers. Determine all functions $f : R^{+} \to R^{+}$ such that for all positive real numbers x and y :

$$f(x + f(xy)) + y = f(x)f(y) + 1$$